

Prime Numbers and the Fundamental Theorem of Arithmetic

- **Exercise 1**

Find all primes p such that $p + 1$ is prime.

- **Solution 1.1**

Suppose p is a prime, then $p > 1$. The only even prime is 2 because if $2 \mid p \implies p = 2k$ for some $k \in \mathbb{N}$. If $k = 1$, then $p = 2$, and if $k \neq 1$, then $p \neq 2$ while $2 \mid p$.

If $p = 2$, then $p + 1 = 3$ is a prime. Now suppose p is a prime with $p > 2$. Then p cannot be even, so p must be odd. It follows that $p + 1$ is even, but since $p + 1 > p > 2$, the integer $p + 1$ cannot be prime. Therefore, the only consecutive prime numbers are 2 and 3.

- **Exercise 2**

(1) Prove that the numbers

$$100! + 2, 100! + 3, \dots, 100! + 100$$

are composite.

- **Solution 2.1**

For each $k \in \mathbb{N}$ with $2 \leq k \leq 100$, we have $k \mid 100!$. We also have $k \mid k$, so $k \mid (100! + k)$. Since $k \neq 1$ and $k \neq 100! + k$, the integer $100! + k$ has a positive divisor different from 1 and itself. This means it is composite. Hence, $\forall k \in \mathbb{N}$ with $2 \leq k \leq 100$, the number $100! + k$ is composite.

(2) Let $N \in \mathbb{N}$. Prove that there exist N consecutive composite numbers.

- **Solution 2.2**

For each $N \in \mathbb{N}$, the following N numbers

$$(N + 1)! + 2, (N + 1)! + 3, \dots, (N + 1)! + N + 1$$

are consecutive composite numbers.

- **Exercise 3**

Suppose $n \in \mathbb{N}$, and p is a prime number. Prove that n^2 is divisible by p if and only if n is divisible by p .

- **Solution 3.1**

($\implies$) Suppose $p \mid n^2$. By Corollary 1.4(1), since $n^2 = n \cdot n$, we have $p \mid n \vee p \mid n \implies p \mid n$.

($\impliedby$) If $p \mid n \implies \exists k \in \mathbb{N}$ such that $n = kp$, then $n^2 = k^2 p^2 = p(k^2 p) \implies p \mid n^2$.

- **Exercise 4**

Prove that a positive integer is a perfect square if and only if every exponent in its prime factorization is even.

- **Solution 4.1**

($\implies$) Suppose $n \in \mathbb{N}$ and there exists $a \in \mathbb{N}$ such that $n = a^2$. Consider the prime factorization of a . By the fundamental theorem of arithmetic, there exist distinct prime numbers $p_1, p_2, \dots, p_r$ and exponents $\alpha_1, \alpha_2, \dots, \alpha_r \in \mathbb{N}$ such that $a = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_r^{\alpha_r}$. Then

$$\begin{aligned} n &= a^2 = (p_1^{\alpha_1} p_2^{\alpha_2} \dots p_r^{\alpha_r})^2 \\ &= p_1^{2\alpha_1} p_2^{2\alpha_2} \dots p_r^{2\alpha_r}. \end{aligned}$$

By the uniqueness of the fundamental theorem of arithmetic, this expression represents the prime factorization of n , where every exponent is even.

($\impliedby$) Suppose the prime factorization of n has even exponents, that is, $n = p_1^{2\alpha_1} p_2^{2\alpha_2} \dots p_r^{2\alpha_r}$. Then

we can write $n = (p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r})^2$. Since $p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r} \in \mathbb{N}$, n is a perfect square.

- **Example 5**

Let x, y be two positive integers. If $x \cdot y$ is a perfect square and x itself is a perfect square, prove that y is a perfect square.

- **Solution 5.1**

By Exercise 4, a positive integer is a perfect square if and only if every exponent in its prime factorization is even.

By Proposition 2.4, for any prime p , we have

$$\mathcal{U}_p(xy) = \mathcal{U}_p(x) + \mathcal{U}_p(y).$$

Since xy and x are perfect squares, $\mathcal{U}_p(xy)$ and $\mathcal{U}_p(x)$ are even integers. Since the difference between two even integers is even, $\mathcal{U}_p(y) = \mathcal{U}_p(xy) - \mathcal{U}_p(x)$ must be even. Thus, every exponent in the prime factorization of y is even, which implies y is a perfect square.

- **Exercise 6**

Prove that the remainder obtained by dividing a prime number by 30 is either 1 or a prime number.

- **Solution 6.1**

Let p be a prime number. Dividing p by 30 according to the division algorithm yields

$$p = q \cdot 30 + r \quad \text{with} \quad 0 \leq r < 30.$$

We want to prove that $r = 1$ or r is a prime number. We consider the possible cases for r :

- If $r = 0$, then $p = 30q$. Since p is prime, this implies $p = 30$ and $q = 1$, but 30 is not a prime number, which is a contradiction.
- Suppose $r > 0$ and r is composite. Then r must have a prime factor s such that $s \leq \sqrt{r} < \sqrt{30} < 6$. Thus, $s \in \{2, 3, 5\}$. Since s is a prime factor of 30, we have $s \mid 30$, and since $s \mid r$, it follows that $s \mid (30q + r)$, which means $s \mid p$. Since p is prime, we must have $s = p$. This implies $p \in \{2, 3, 5\}$. However, if $p \in \{2, 3, 5\}$, then dividing p by 30 yields a remainder $r = p$, which is a prime number, contradicting the assumption that r is composite.

Therefore, the remainder r cannot be 0 or a composite number, meaning it must be either 1 or a prime number.